Angewandte Chemie
Eine Zeitschrift der Gesellschaft Deutscher Chemiker

Supporting Information

© Wiley-VCH 2008

69451 Weinheim, Germany

SUPPLEMENTARY INFORMATION

S-Adenosyl-L-methionine:hydroxide adenosyltransferase, a SAM enzyme **

Hai Deng, Catherine H. Botting, John T.G. Hamilton, Rupert J. M. Russell, David O'Hagan[*,]

1. Purification of S-Adenosyl-L-methionine:hydroxide adenosyltransferase

The clone containing ORF PH0463 from the marine archaea *P. horikoshii* OT3 was purchased from NBRC (NITE-DOB, Japan) (NBRC no: NBRC G01-000-971, Clone ID: S2_2219, EMBL accession No. BAA29549). The *PH0463* ORF was subcloned into pET28a(+) vector (Novagen) using the following two primers: PH0463-F(GCAGGAGGAATT*CATATG*ATAACCTTAACCACCGACTTCGG) and PH0463-R: (CGCCG*CTCGAG*TCATAGCAACCTCACCCTTAGCTC) with restriction sites NdeI and XhoI, respectively. The resultant plasmid pET-PH0463 was transferred into *E coli* BL21 C43(DE3) competent cell and grown in Luria broth containing kanamycin (100 μg/ml) at 37°C until an absobance of 0.6 at 600 nm was reached. The PH0463 protein was over-expressed in *E. coli* by adding IPTG (1mM) and cells were left to grow at 16^{0}C for 16 h. Cells were then harvested by centrifugation and were subject to sonication to induce lysis. The cell-free extract with PBS and imidazole (10 mM) was then centrifuged twice at 20,000g at 4^{0}C for 15 min. The supernatant was collected by passing through a Ni-affinity column (Qiagen) and the active fractions were eluted by adding PBS buffer with imidazole (100 mM). The eluent was dialysed for 16h at 25°C by adding thrombin (0.5 unit; Sigma-Aldrich Co. Ltd.,) and adenosine deaminase (0.01 unit, Sigma-Aldrich Co. Ltd.,). The dialysate was then passed through a Ni-affinity column for a second time and the eluent was subjected then to size exclusion chromatography (Size Exclusive column -Amersham Co). The active fractions were collected and subjected to enzymatic assay.

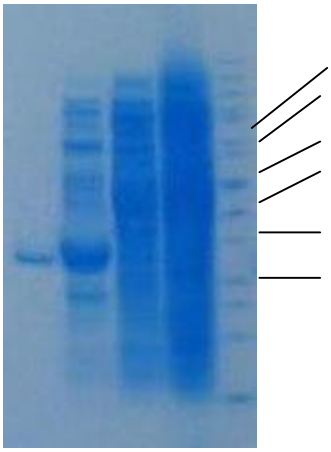


Figure 1. SDS-Page of the progressive purification of S-adenosyl-L-methionine:hydroxide adenosyltransferase. 1. Protein molecular weight markers; 2. Cell-free extract; 3. Eluent after the first Ni-affinity column; 4. Eluent after the second Ni-affinity column; 5. Eluent after size exclusion chromatography. The identity of the protein was confirmed by MS-MS mass spectrometry (Applied Biosystems 4800 TOF/TOF analyser) of a trypsic digest of the purified protein.

2. Enzyme Assay

In order to determine the pH optimum of the enzyme activity was assayed in $NaHCO_3$-citric acid buffer (100mM, pH 4.5), phosphate buffer (100mM, pH 7.2), Tris-HCl buffer (100 mM, pH 8.5), Na_2CO_3-$NaHCO_3$ buffer (100mM, pH 10), CAPS-NaOH (100mM, pH 11) and CAPS-NaOH (100mM, pH 12.5) buffer.

Experiments to determine the kinetic parameters involved incubation of the enzyme (0.2-0.8 μM), SAM (10 μM-1 mM) and 5'-ClDA (100 μM – 3 mM) in Tris-HCl buffer (100mM, pH 8.5), in a final volume of 100 μL. The enzymatic reaction was quenched by adding ice-cold ethanol (500 μl). After freeze-drying the resultant solution, the residue was dispersed in water (100 μl) and an aliquot was subject to analytical HPLC assay. For SAH **6**

inhibition experiments, SAH **6**(50 µM), enzyme (0.2 µM) and the indicated SAM concentration in Tris-HCl buffer (100mM, pH 8.5), at a final volume of 100µL.

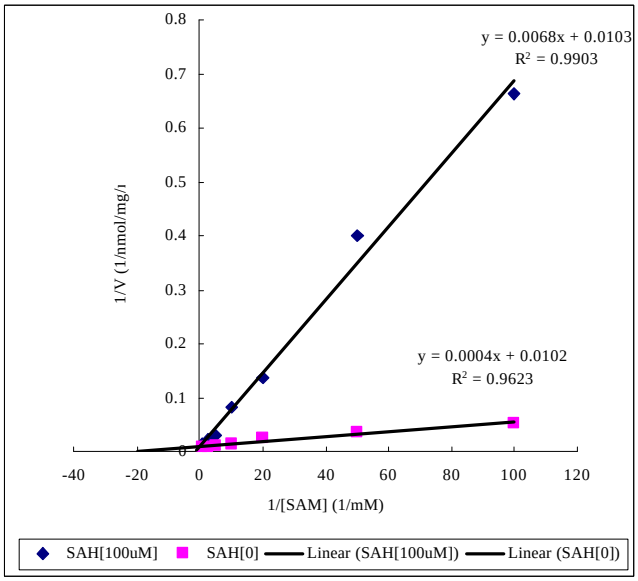


Figure 2. Double recriprocal plot showing SAM as a substrate and the inhibition of SAM conversion with SAH. Inhibition kinetic studies of PH0463 protein with SAM (0.01 mM - 1 mM) as the substrate and SAH as an inhibitor of SAM conversion. K_m for SAM = 39.2 µ? , k_{cat} = 0.14 s^{-1}

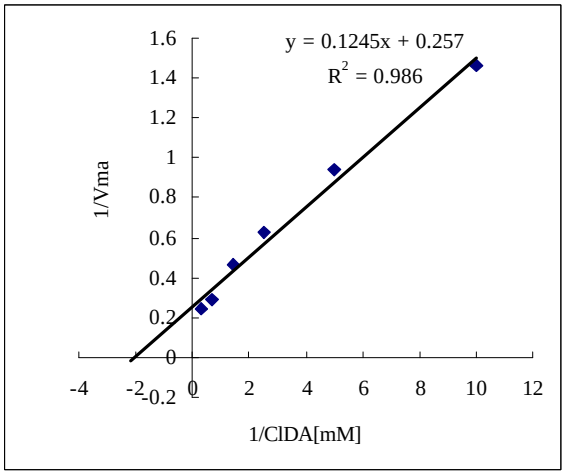
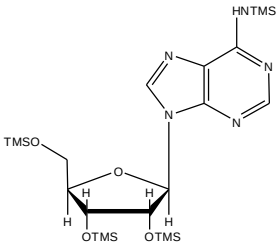


Figure 3. Double recriprocal plot showing 5'-ClDA **3** (0.1 mM to 3 mM) as a substrate for the enzyme. K_m for 5'-ClDA = 410 µM, k_{cat} of 0.31 min^{-1}

3. GC-MS method for measurement for the incorporation of the ^{18}O-isotope from [^{18}O]-water into [5'-^{18}O]-adenosine.

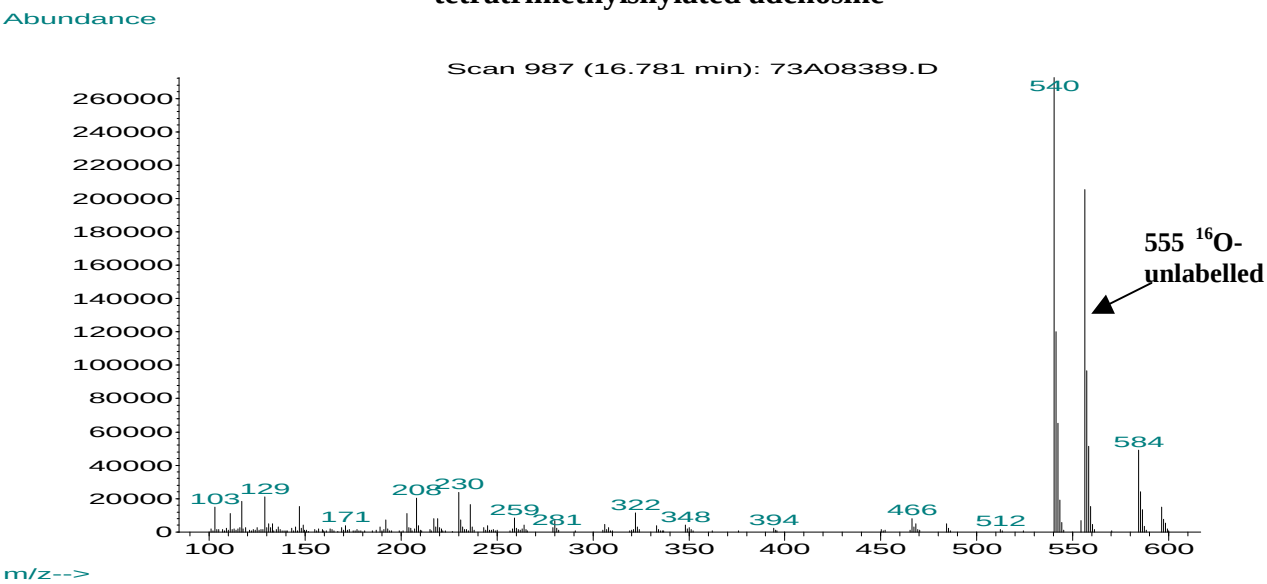
The enzyme (10µM, 60 µL) was incubated with $H_2{}^{18}O$ (95 atom%, Sigma-Aldrich Co., 100 µL) and an aliquot (10 µL) of a 20mM SAM solution in PBS buffer at 37 ^{o}C for 16 h. After addition of ethanol (500µL) the solution was freeze-dried and the residue was treated with an excess of N-methyl-N-(trimethylsilyl) trifluoroacetamide and then heated for 60 min at 100°C. GC-MS analysis was performed on an Agilent 5890 GC instrument which was directly attached to Agilent 5973A mass selective detector (MSD). The GC was equipped with an Ultra 1 fused-silica capillary column (Agilent Technologies; 12m × 0.25mm × 0.17µm). The oven temperature was programmed to hold for 1 min at 100°C and then ramp at 10°C/min to 300°C. The injector and transfer line temperatures were set at 250°C and the per-trimethylsilylated sample (1µl) automatically injected in the splitless

mode. The MSD was operated in the selected ion monitoring mode with positive chemical ionization using methane as the reagent gas. Ion currents were measured at *m/z* 555, 556 and 557 amu. Percentage incorporation of ^{18}O into adenosine was calculated at 48% by the formula [(integral of ion current at *m/z* 557/sum of integrals of ion currents at *m/z* 555 and 557) x 100] after ion current at *m/z* 557 had been corrected for natural abundance. The latter correction was applied by measuring the natural abundance of this ion in an authentic standard at the time of analysis. The diluation from 95-atom % to 48-atom-% can be accounted for by assay dilution.



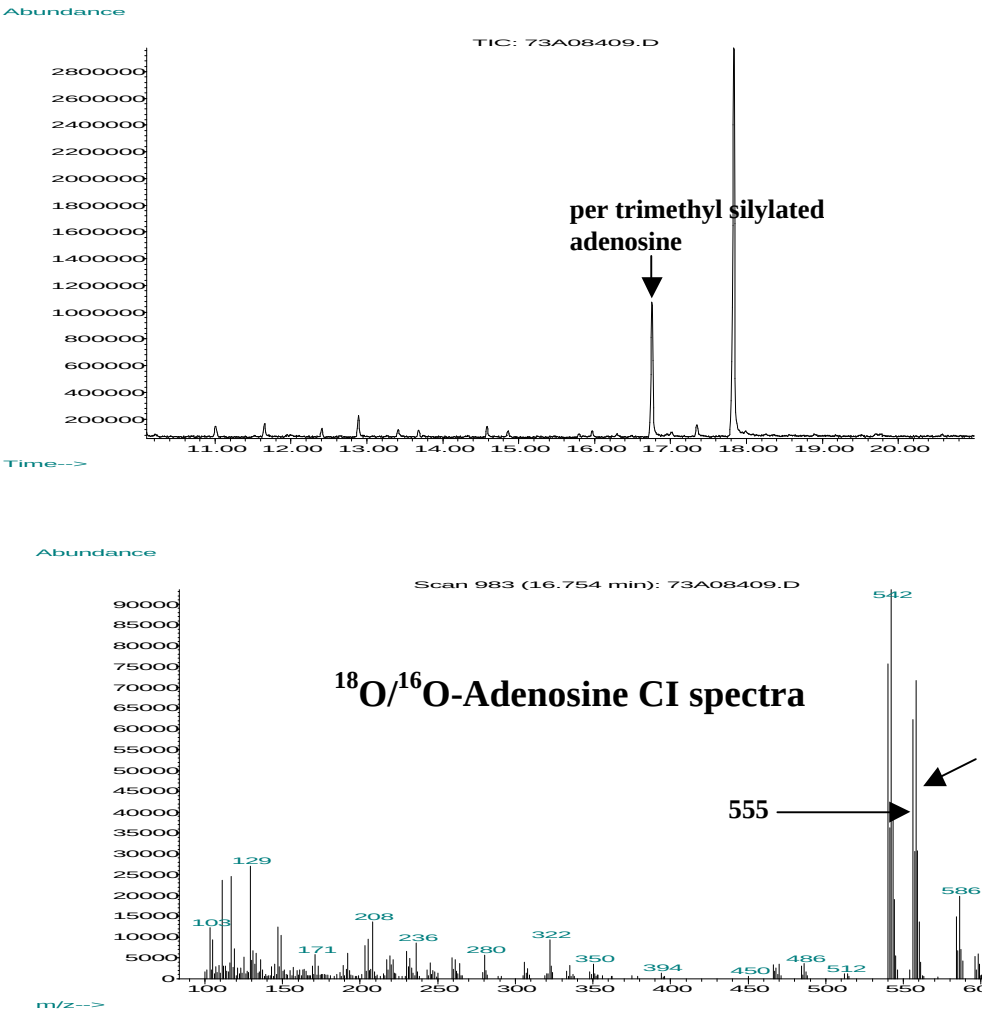
mass 555amu

tetratrimethylsilylated adenosine



CI-Mass spectrum of tetratrimethysilylated adenosine (unlabelled).

CI TIC ^{18}O sample



Upper: GC trace/retention time of tetratrimethylsilylated adenosine

Lower: CI-mass spectrum of labelled adenosine recovered from an enzyme reaction carried out in $^{18}H_2O$ (48 atom %)

4. MS-MS methods

The gel band was excised and cut into 1mm cubes. These were then subjected to in-gel digestion, using a ProGest Investigator in-gel digestion robot (Genomic Solutions, Ann Arbor, MI) using standard protocols. Briefly the gel cubes were destained by washing with acetonitrile and subjected to reduction and alkylation before digestion with trypsin (36 ng) at 37°C. The peptides were extracted with 10% formic acid and concentrated down to 20 μL using a SpeedVac (ThermoSavant).

For analysis by nanoLC-ESI MS-MS, they were then separated using an UltiMate nanoLC (LC Packings, Amsterdam) equipped with a PepMap C18 trap & column, using a 30 min gradient of increasing acetonitrile concentration, containing 0.1 % formic acid (5-35% acetonitrile in 18 min, 35-50% in a further 7, followed by 95% acetonitrile to clean the column). The eluent was sprayed into a Q-Star Pulsar XL tandem mass spectrometer (Applied Biosystems, Foster City, CA) and analysed in Information Dependent Acquisition (IDA) mode. MS/MS data for doubly and triply charged precursor ions was converted to centroid data, without smoothing, using the Mascot Daemon 2.1 (Matrix Science, London) data import filter for Sciex Analyst. The MS/MS data file generated was analysed using the Mascot search engine against an internal database containing the predicted sequence. The data was searched with tolerances of 0.2 Da for the precursor and fragment ions, trypsin as the cleavage enzyme, one missed cleavage, carbamidomethyl modification of cysteines as a fixed modification and methionine oxidation (and ^{18}O incorporation where appropriate) selected as a variable modification.

For MALDI MS and MS-MS, the sample solution (0.5 μL) was applied to the MALDI target along with α-cyano-4-hydroxycinnamic acid matrix (0.5 μL, 10 mg/mL in 50:50 acetonitrile:0.1% TFA). MALDI MS and MSMS was acquired using a 4800 MALDI TOF/TOF Analyser (Applied Biosystems, Foster City, CA) equipped with a Nd:YAG 355 nm laser and calibrated using a mixture of peptides. The data was processed as described above with tolerances of 100 ppm for the precursor and 0.5 Da for the fragment ions.

Investigating Oxygen-18 incorporation into the enzyme at Asp-68.

MS peptide sequences: GTVHVGVIDPGVGTER

Peptide aa 60-75 (triply charged)

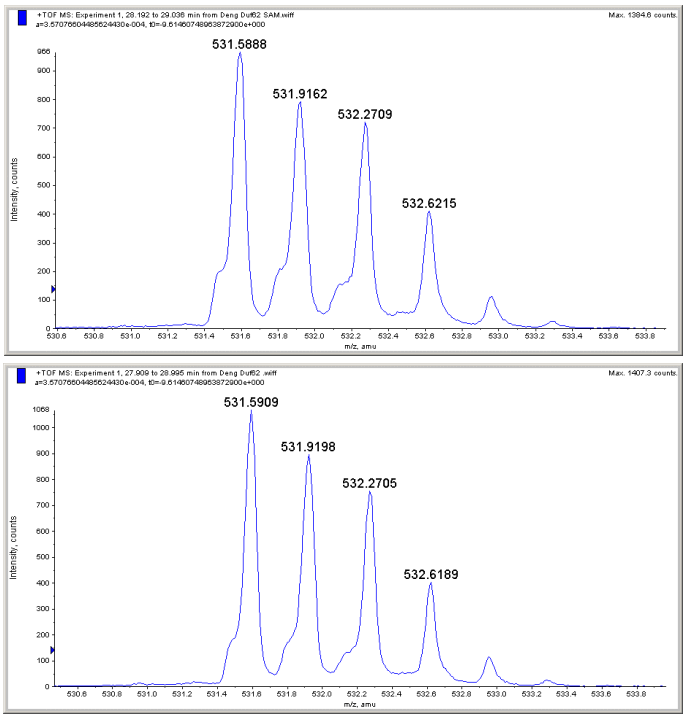


Figure 4 - Mass analyses of the molecular ions of triply charged tryptic peptide corresponding to amino acids 60-75 (GTVHVGVIDPGVGTER) did not show evidence for the incorporation of ^{18}O from $H_2{}^{18}O$ (95-atom-%) at Asp-68 in the presence (upper) or absence (lower) of SAM.

The enzyme (20µM, 90µL) was incubated with 100 µL $H_2{}^{18}O$ (95 atom %, Sigma-Aldrich Co.) and 10 µL of 20mM SAM in PBS buffer overnight. An aliquot (40 µL) was subject to HPLC analysis, and simultaneously an aliquot (50µL) was analysed by SDS-PAGE and the resultant protein band excised and subjected to trypsin in-gel digestion (see above). The resultant peptides were then separated using an UltiMate nanoLC (LC Packings, Amsterdam) equipped with a PepMap C18 trap & column, using a 30 min gradient of increasing acetonitrile concentration, containing 0.1 % formic acid (5-35% acetonitrile in 18 min, 35-50% in a further 7, followed by 95% acetonitrile to clean the column). The eluent was sprayed into a Q-Star Pulsar XL tandem mass spectrometer (Applied Biosystems, Foster City, CA) and analysed in Information Dependent Acquisition (IDA) mode.

MS peptide sequences: GTVHVGVIDPGVGTER

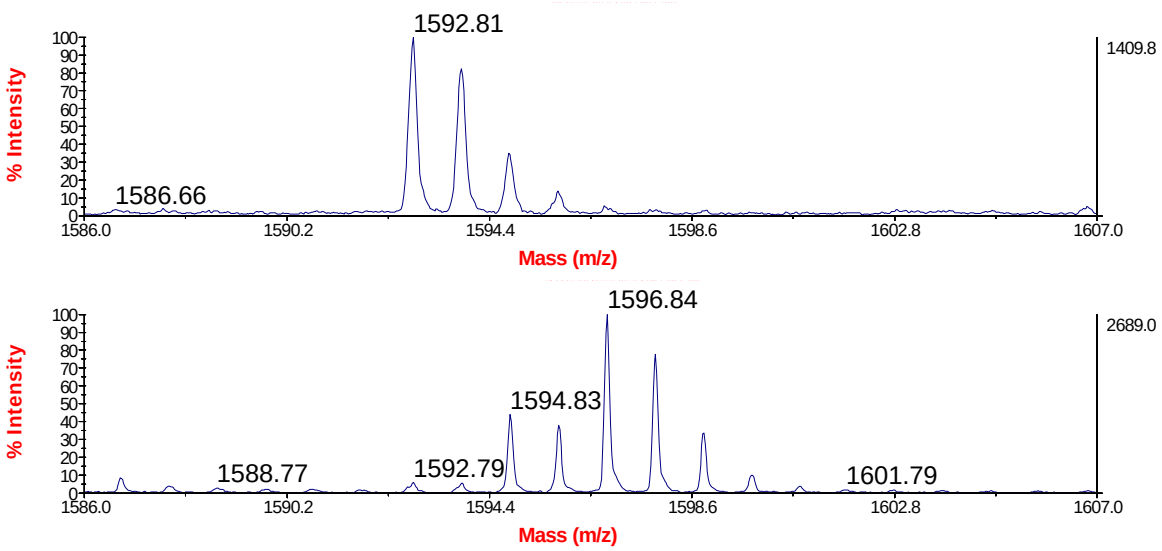


Figure 5 - Control experiment. Oxygen-18 incorporation into a protein fragment of the enzyme as assessed by enzymatic digest in $H_2{}^{18}O$ – MALDI mass spectrum of tryptic peptide corresponding to amino acids 60-75 performed in (Upper) $H_2{}^{16}O$ and (Lower) $H_2{}^{18}O$.

A sample of the enzyme was subjected to manual in-gel digestion, where the destained, reduced and alkylated sample was incubated in $H_2{}^{18}O$ ((95-atom-%, 15 µL), 1 M ammonium bicarbonate (0.3 µL), trypsin (0.3 µL of 0.2 µg/µL) in 0.01% formic acid at 37°C for 16h. The sample solution (0.5 µL) was applied to the MALDI target along with α-cyano-4-hydroxycinnamic acid matrix (0.5 µL, 10 mg/mL in 50:50 acetonitrile:0.1% TFA). MALDI-MS and MS-MS was acquired using a 4800 MALDI TOF/TOF Analyser (Applied Biosystems, Foster City, CA). The lower spectrum shows a 95% incorporation of ^{18}O.

5. Alignment information for Figure 1

Sample Name	Protein	Organism	NCBI access N
I-fluorinase	5'-fluoro-5'-deoxy-adenosine synthase (fluorinase)	*Streptomyces cattleya*	46015529
II-chlorinase	Duf62 protein	*Salinispora tropica-CNB440*	126466090
III	Duf62 protein	*Acidobacteria bacterium* Ellin345	83814680
IV	Duf62 protein	*Salinibacter ruber* DSM 13855	83814680
V	Duf62 protein	*Prosthecochloris aestuarii* DSM 271	68551507
VI	Duf62 protein	*Thermotoga maritima* MSB8	15643425
VII	Duf62 protein	*Listeria monocytogenes* EGD-e	16804625
VIII	Duf62 protein	*Chlorobium ferrooxidans* DSM 13031	110596904
IX	Duf62 protein	*Staphylothermus marinus* F1	126466090
X	Duf62 protein	*Pyrococcus horikoshii* OT3	14590374
XI	Duf62 protein	*Escherichia coli O157:H7*	15804403
XII	Duf62 protein	*Staphylococcus aureus subsp. aureus*	57651078

7. Protein database (PDB) codes for the crystallographic data for the four Duf-62 proteins that have had their structures solved by crystallography.

Duf 62 protein from *Methanococcus Jannaschii* DSM 2661	PDB: 2f4n
Duf 62 protein from *Pyrococcus horikoshii* OT3	PDB: 1WU8
Duf 62 protein from *Thermus Thermophilus* Hb8	PDB: 2CW5
Duf 62 protein from *Thermotoga maritima*	PDB: 2ZBU

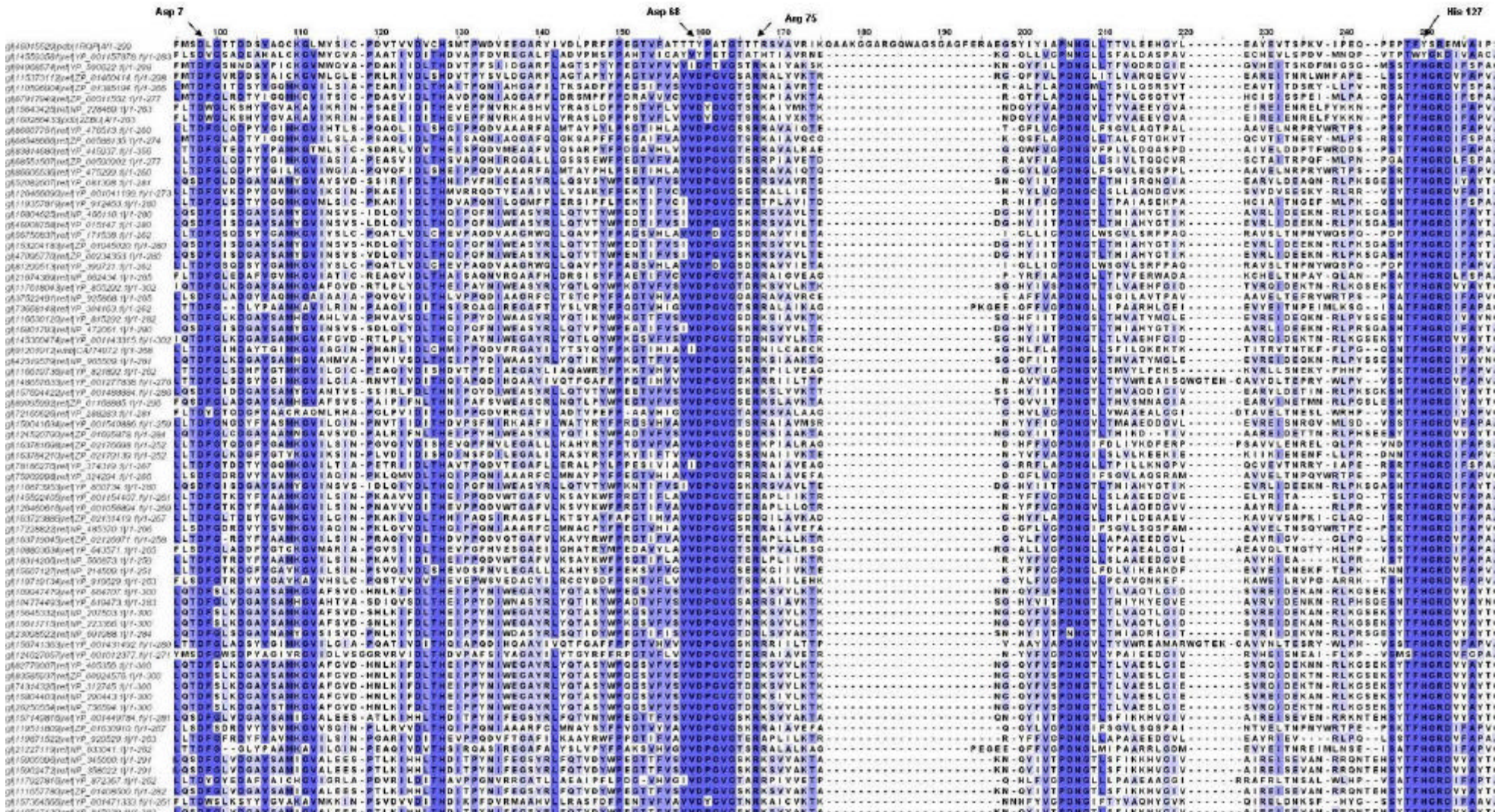

Asp 7
Asp 68
Arg 75
His 127

Figure 6. Amino acid sequences of around ninety Duf-62 proteins,derived from genome sequence data. The sequences are compared with the fluorinase and the chlorinase (entries 1 and 2 respectively). The arrows point to the three active site residues (His, Arg and Asp) which are hydrogen bonded together (see Figure 3 in the text), and also the Asp residue which anchors the 2'- and 3'- hydroxyl groups of the ribose ring in adenosine/SAM. These residues are highly conserved through the Dug-62 family of enzymes. The numbering of the arrowed residues is for the *P. horikoshi* enzyme. It isclear that the although the chlorinase and fluorinase enzymes have some similarity to the Duf-62 family, they do not retain the hydrogen bonded His, Arg and Asp triad, and as a consequence are fundamentally different.

[End]